

Level 3: Advanced Topics

Specialised methods for complex clinical data: workshops and consultancy

Organisation: Neudata · #ClearDataClearImpact

> **Status:** outline. Modules are developed on request, after Level 1 and Level 2 have run.

1. What Level 3 is

Real clinical data are often more complex than the cohort used in Levels 1 and 2: repeated measurements on the same patients, patients clustered in wards or hospitals, competing outcomes, large routine registries, or questions about prediction and diagnosis rather than association. Level 3 offers specialised modules, each tied to the kind of data or question participants meet in their own work. Neudata runs them as workshops and consultancy: a short, intensive teaching block followed by hands-on work on the participant's own project, with Neudata support until a defensible analysis plan or analysis is in place.

2. Format

Who	Level 2 graduates (or researchers with equivalent R and regression experience), usually 3–10 per workshop; individual researchers or project teams for consultancy.
Prerequisites	Level 2 (or equivalent): reproducible R projects, data cleaning, linear and logistic regression with diagnostics, multiple imputation. A real question or dataset is strongly recommended.
Workshop format	1–2 sessions of 3 hours: concepts → worked example in R → participants apply it to their own data.
Consultancy format	A scoping meeting, then 2–4 follow-up meetings over 1–3 months; the participant does the analysis, Neudata reviews the plan, code and interpretation.
Software	R and RStudio; module-specific packages (named per module).
Output	A statistical analysis plan, a reviewed analysis script and draft Methods/Results text for the participant's project, or a structured appraisal for the appraisal-focused modules.
How to request	The department or a project team contacts Neudata with a short description of the question and data; Neudata proposes the module(s) and the format.

3. The modules

Module A: Systematic review and meta-analysis

Aims. Participants will be able to (1) write and register (PROSPERO) a systematic-review protocol; (2) build a reproducible multi-database search and screen records in pairs; (3) extract data and assess risk of bias (RoB 2, ROBINS-I); (4) pool effect estimates with fixed- and random-effects models, interpret heterogeneity (I^2 , prediction intervals) and draw forest and funnel plots; and (5) report to PRISMA 2020 and grade certainty (GRADE).

Prerequisites. Level 2; a review question (PICO) agreed with Neudata. Packages: meta, metafor.

Session	Content	Product
A1: Protocol, search and screening	Review question and eligibility; PROSPERO; search strategy across databases; dual screening; PRISMA flow diagram; data-extraction form	Registered (or ready-to-register) protocol and search strategy
A2: Bias, pooling and reporting	Risk-of-bias assessment; pooling ORs/RRs/mean differences in R; fixed vs random effects; I^2 , subgroups and meta-regression (introduced); small-study effects and funnel plots; GRADE; PRISMA 2020	Forest plot and draft Results for the participant's review

Module B: Prediction models and machine learning

Aims. Participants will be able to (1) distinguish prediction from causal questions; (2) calculate the sample size needed to develop a model; (3) develop a logistic or survival prediction model, assess discrimination and calibration, and validate it internally (bootstrapping); (4) understand what machine-learning methods add and where they mislead (data leakage, overfitting, fairness); and (5) report with TRIPOD+AI and appraise with PROBAST.

Prerequisites. Level 2 (logistic regression in depth). Packages: rms, pmsampsize, pROC, glmnet, tidymodels (introduced).

Session	Content	Product
B1: Building a model properly	Prediction vs causation; candidate predictors; sample size for prediction models; development on FLUID-ICU or own data; discrimination and calibration plots; bootstrap internal validation; presenting the model (nomogram, risk chart)	Model report with optimism-corrected AUC and a calibration plot
B2: Machine learning and appraisal	Penalised regression and a tree-based method compared with logistic regression; data leakage; external validation and updating; fairness; appraising a published AI/ML model with TRIPOD+AI and PROBAST	Structured appraisal of a published model, or a validation plan for one's own model

Module C: Diagnostic-accuracy studies

Aims. Participants will be able to (1) design a diagnostic-accuracy study (index test, reference standard, spectrum, blinding, verification); (2) calculate sensitivity, specificity, predictive values and likelihood ratios with 95% CIs, and draw ROC curves with AUC; (3) choose and justify a cut-off; (4)

compare two tests on the same patients; and (5) report to STARD 2015 and appraise with QUADAS-2.

Prerequisites. Level 2. Packages: epiR, pROC.

Session	Content	Product
C1: Design, analysis and reporting	Diagnostic questions and designs; spectrum, verification and incorporation bias; accuracy table with CIs; predictive values across prevalences; ROC/AUC and cut-offs (e.g. lactate for 28-day mortality); comparing two tests (paired ROC); STARD 2015 and QUADAS-2	Completed accuracy analysis and a STARD-based study outline
C2 (optional): Your own diagnostic question	Protocol for a unit-level accuracy study; sample size for sensitivity and specificity; data-collection form	Protocol outline

Module D: Advanced survival analysis

Aims. Participants will be able to (1) fit and check Cox models (proportional hazards, functional form, influential points); (2) use time-varying covariates and avoid immortal-time bias; (3) analyse competing risks (cumulative incidence functions, cause-specific and Fine–Gray models), for example ICU discharge alive vs death; (4) present restricted mean survival time as an alternative to the HR; and (5) report time-to-event analyses clearly.

Prerequisites. Level 2; the Kaplan–Meier and Cox basics from Level 1 (FLUID-ICU Question 1). Packages: survival, survminer, tidycmprsk, cmprsk.

Session	Content	Product
D1: Beyond the basic Cox model	Checking proportional hazards; stratification and time-varying effects; time-varying covariates and the counting-process format; immortal-time bias and landmark analysis; restricted mean survival time	Cox analysis with PH checks and a time-varying covariate
D2: Competing risks	Why 1 – KM overestimates risk with competing events; cumulative incidence; cause-specific vs Fine–Gray models and when to use each; plotting and reporting	Competing-risk analysis of FLUID-ICU or own data

Module E: Longitudinal data and multilevel (mixed) models

Aims. Participants will be able to (1) structure repeated-measures data (long vs wide) and plot individual trajectories; (2) fit linear and logistic mixed-effects models with random intercepts and slopes; (3) interpret fixed effects, random effects and intraclass correlation; (4) handle missing data and dropout in repeated measures; and (5) know when generalised estimating equations (GEE) are the better choice.

Prerequisites. Level 2 (linear and logistic regression in depth). Packages: lme4, glmmTMB, geepack, ggeffects.

Session	Content	Product
E1: Repeated measures over time	Long vs wide data; spaghetti plots; why ordinary regression is wrong for repeated measures; linear mixed models for daily measurements (e.g. fluid balance or SOFA on days 1–3); random intercepts and slopes; time as a predictor	Mixed-model analysis of a repeated outcome with a Results paragraph
E2: Clustered outcomes and GEE	Binary repeated or clustered outcomes; logistic mixed models; GEE and population-average vs subject-specific effects; ICC; model checking; reporting	Comparison of mixed-model and GEE estimates for one question

Module F: Complex data: clustered designs, repeated measures and large registries

Aims. Participants will be able to (1) recognise clustering (patients within wards, ICUs or hospitals) and its consequences for standard errors and sample size; (2) analyse cluster-randomised and multicentre data (centre effects, design effect); (3) work with large routine datasets and registries (data linkage, data quality checks at scale, coding systems such as ICD, efficient data handling in R); and (4) plan causal analyses of observational data (DAGs, propensity scores, the target-trial idea, introduced).

Prerequisites. Level 2; Module E recommended. Packages: data.table or arrow, MatchIt, WeightIt, cobalt, dagitty.

Session	Content	Product
F1: Clustered and multicentre data	Design effect and ICC; centre as fixed vs random effect; cluster-robust standard errors; analysing a cluster-randomised or multicentre study; reporting (CONSORT cluster extension)	Analysis plan for a clustered or multicentre dataset
F2: Large registries and routine data	Data linkage and de-identification; data quality at scale; coding systems; efficient data handling; causal questions from routine data (DAGs, propensity scores with matching, weighting and balance checks, and target-trial emulation, introduced); RECORD reporting	Data-management and analysis plan for a registry-based question

4. Choosing a module

If you want to...	Module
Summarise all the evidence on a question	A: Meta-analysis
Build or judge a risk score, prediction tool or AI model	B: Prediction and ML
Test how good a test, score or sign is at detecting a condition	C: Diagnostic accuracy
Analyse time-to-event outcomes with competing events or changing exposures	D: Advanced survival

If you want to...	Module
Analyse measurements repeated over time on the same patients	E: Longitudinal and mixed models
Analyse multicentre, cluster-randomised or registry data	F: Complex data

Key free resources for each module are listed in `Course/References/further_reading.md`: the Cochrane Handbook and PRISMA 2020 (A); TRIPOD+AI and the Riley sample-size paper (B); STARD 2015 and QUADAS-2 (C); *Statistics at Square One* and the Bland–Altman Statistics Notes (D, E); the CONSORT cluster extension (F).